0739. Daily Temperatures

Medium · Stack

leetcode.com/problems/daily-temperatures

Problem

- For each day, how many days until a warmer temperature? 0 if never.
- Example: [73,74,75,71,69,72,76,73] → [1,1,4,2,1,1,0,0].

Intuition

- Keep a stack of days still waiting for a warmer day; their temperatures are decreasing from bottom to top.
- A new warmer day resolves every waiting day on top that is colder than it.

Brute Force

- For each day scan forward: O(n^2).

Optimal Approach

- For i: while stack non-empty and T[i] > T[top]: j = pop, ans[j] = i - j.
- Push i.

Worked Example

```
[73,74,75,71,69,72,76,73].

i=0 73 push              st=[0]
i=1 74 pop 0 ans=1       st=[1]
i=2 75 pop 1 ans=1       st=[2]
i=3 71 push              st=[2,3]
i=4 69 push              st=[2,3,4]
i=5 72 pop 4(1) pop 3(2)  st=[2,5]
i=6 76 pop 5(1) pop 2(4)  st=[6]
i=7 73 push              st=[6,7]  -> leftovers stay 0
```

Java Solution

```java
public class DailyTemperatures {
    public int[] dailyTemperatures(int[] t) {
        int[] ans = new int[t.length];
        Deque<Integer> st = new ArrayDeque<>();
        for (int i = 0; i < t.length; i++) {
            while (!st.isEmpty() && t[i] > t[st.peek()]) {
                int j = st.pop();
                ans[j] = i - j;
            }
            st.push(i);
        }
        return ans;
    }
}
```

Complexity

- Time: O(n), each index pushed/popped once. Space: O(n).

Pitfalls

- Strictly warmer: use >, not >=.